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Palaeomagnetic Studies in the British Caledonides—II The Younger Gabbros of Aberdeenshire, Scotland

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Summary

A palaeomagnetic survey of the Younger Gabbros of north-east Scotland is presented. Forty-one sites have been sampled from seven gabbro masses (Belhelvie, Arnage, Maud, Insch, Huntly, Boganclogh and Haddo House) of which the NRM at 37 of these sites is significantly grouped. The bulk of the collection shows high stability on AF and thermal cleaning and 32 sites have significant direction after cleaning and yield a stable direction $D = 168°$, $I = +34°$ ($k = 9{\cdot}7$). All sites have the same polarity. A palaeomagnetic north pole is derived at 9° E, 11° S which is 16° away from a previous estimate for the gabbros based on total NRM only. A fold test has been applied to the igneous lamination of three of the bodies and it is concluded that the bodies were subjected to local F_3 folding above their Curie point which, in most of the samples tested, falls in the range 500–600 °C. The Rb–Sr and K–Ar data for the gabbros are reviewed and the best estimate of the magnetic age is approximately 470 My.

1. Introduction

The basic intrusions of north-east Scotland form a major igneous province and consist of several large and many small intrusions of predominantly gabbroic composition. The outcrop covers an area of some 2500 km^2, and the larger bodies are the Huntly, Insch, Haddo House, Arnage, Maud and Belhelvie masses (Fig. 1). They predate the post-orogenic granites of the area and appear to have been intruded at about the climax of Caledonian regional metamorphism. They have been divided by Read (1919) into two suites: an 'Older' suite of amphibolitized and foliated bodies and a 'Younger' unmetamorphosed suite. The Younger suite, which forms the subject of the present paper, has been extensively examined by Read (1923, 1935, 1955, 1956), Read & Farquhar (1952, 1956) and Stewart (1946), but their interpretation of structural relationships has been a matter of controversy. Vertical rhythmic banding occurs in the Belhelvie mass and steeply dipping bands occur in the Huntly and Insch masses. These allow an estimate of thickness, and Clarke & Wadsworth (1970) calculate that the Insch mass is as much as 6 km thick. On the assumption that the banding is due to crystal accumulation under gravity,

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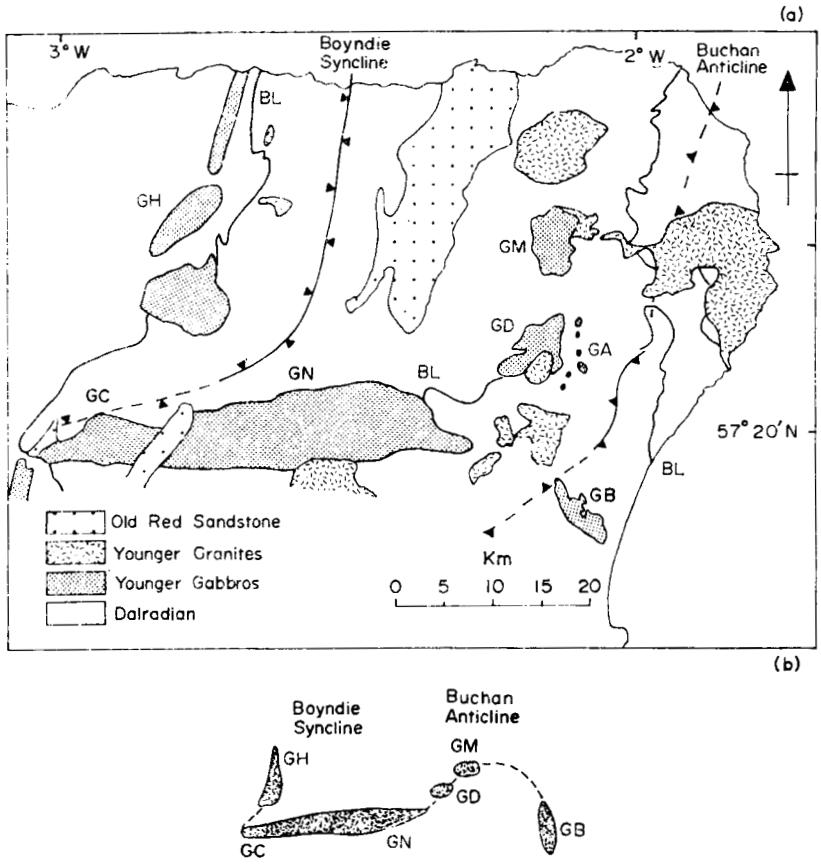


FIG. 1(*a*) Simplified geological map of north-east Scotland showing the positions of the Younger Gabbro complexes. BL is the outcrop of the Boyne Line. The Huntly mass is shown as three separate outcrops following evidence from drilling that Dalradian rocks underlie the intervening drift cover (Munro 1970). (*b*) Structural interpretation of the gabbro masses after Shackleton (in discussion of Read & Farquhar 1956). The masses are labelled as in Table 1.

Stewart (1946) and Shackleton (1948) have proposed that the bodies have been deformed since consolidation by a NNE–SSW folding about the Buchan anticline (Fig. 1). Read found this explanation unacceptable (Read & Farquhar 1956) and argued that deformation of the basic masses would distort the petrographic textures. In addition he believed that the recumbent folding of the Banff Nappe was not separated by a sufficient time interval from the Buchan anticlinal folding for the intrusion of the Younger Gabbros.

In an attempt to resolve this disagreement Blundell & Read (1958) undertook a palaeomagnetic study of six of the Younger Gabbro masses. They showed that these bodies have similar *in situ* directions of total NRM and concluded that they have remained relatively undisturbed since their formation. Stewart & Johnson (1960) challenged this conclusion and suggested that the bodies might have been deformed at a temperature above the Curie points of their ferromagnetic minerals.

Recent magnetic and gravity studies (McGregor & Wilson 1967) have provided additional evidence that the exposed bodies are part of a continuous sheet and support the contention that it has been folded after consolidation. Study of the metamorphic textures around the Insch mass has demonstrated that it was

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intruded during the main metamorphic crystallization in the area (Fettes 1970) and examination of the Morven–Cabrach mass has shown that it belongs to the Younger and not the Older suite as originally thought (Allan 1970). The only tests for the stability of magnetization applied to the collection of Blundell & Read (1958) was an examination of the magnetic components acquired by storage in the Earth's field. The present study has been undertaken to determine the directions of magnetization of the gabbros after examination of its stability by AF and thermal methods. It is then possible to see whether they merit a structural interpretation or provide a reliable direction of the geomagnetic field for the time of intrusion of the gabbro bodies.

2. Geological setting

The Younger Gabbros are intruded into a variety of Dalradian country rocks which have been subjected to several structural and metamorphic events. A time scale for these events has been proposed by Johnson (1962, 1963) and endorsed by the work of Fettes (1970). The sequence they propose is:

(i) First nappe folding (F_1) and metamorphism (M_1),

(ii) Second folding (F_2) and metamorphism (M_2),

(iii) Main regional metamorphism (M_3) and

(iv) Folding of the Buchan Anticline and Boyndie Syncline (F_3) followed by minor folding (F_4).

The Older Gabbros predate the regional metamorphism, M_3; the Younger Gabbros postdate this episode and are intruded near the contact between the Lower Dalradian gneisses (Keith Division) and the Upper Dalradian quartzites and greywackes (Banff Division). This contact crops out as the ' Boyne Line ' of Read & Farquhar (1956) (Fig. 1).

The Younger Gabbros include peridotites, picrites, varieties of gabbro, quartz norites and cordierite norites. The evidence from rhythmic banding has been used by Shackleton (in discussion to Read & Farquhar 1956) to propose that the Younger Gabbros were involved in the large scale folding F_3 which overturned the Belhelvie mass. This hypothesis has been generally accepted by later workers (Stewart & Johnson 1960; Johnson 1963; Fettes 1970).

The radiometric evidence for the age of the Younger Gabbros has been summarized by Pankhurst (1970). Rb–Sr age determinations suggest an age of 486 ± 17 My for the intrusion of the gabbros (using a mean Rb^{87} half life of $4{\cdot}85 \times 10^{10}$ yr) and associated gneisses give a slightly younger age of 479 ± 12 My. K–Ar determinations on micas define an isotopic event at 461 ± 5 My and Pankhurst has argued that this represents a cooling age when the gabbros would have been near 150–200 °C. The bodies thus appear to have remained at high temperatures for a prolonged interval after their emplacement.

3. Results

3.1 *Field sampling*

In the course of the present work 41 sites were sampled using a portable drill (Doell & Cox 1967). Cores were oriented by sunsights and/or magnetic bearings, and a clinometer. The number of sites sampled from each gabbro mass is: Huntly (10), Insch (13) and Boganclogh (3), Haddo House (1), Arnage (3), Maud (2), and Belhelvie (9). The maximum thickness of any body sampled is about 4 km for the Belhelvie mass.

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Table 1

Summary of in situ *palaeomagnetic directions*

Site No.	National Grid Reference	No. of cores	N (specimens)	R	k	α_{95}	D	I	Peak Alt.* Field (Oe)	N (specimens)	R	k	α_{95}	D	I	Virtual Geomagnetic North Pole °E	Virtual Geomagnetic North Pole °N
(i) Belhelvie Mass																	
GB 1	945181	5	5	4·45	7·2	30·6	238	+49	500	4	3·92	37·7	15·2	201	+36	338	−11
GB 2	944180	5	5	4·00	4·0	44·0	185	+45	200	5	4·75	16·2	19·6	179	+25	359	−20
GB 3	947173	5	5	4·58	9·6	26·1	27	+50	200	5	4·16	4·7	39·3	163	+46	13	−5
GB 4	942187	5	3	2·63	5·3	59·7	20	+47	300	4	3·86	21·0	20·6	193	+13	344	−25
GB 5	921191	5	2	2·00	239·9	16·2	167	+33	300	3	2·99	231·7	8·1	169	+34	8	−14
GB 6	919194	5	3	—	—	—	—	—	300	5	4·99	543·8	3·3	168	+31	10	−15
GB 7	927186	5	3	2·88	16·3	31·6	174	+64	700	5	4·96	105·4	7·5	180	+39	352	−11
GB 8	929177	5	3	2·94	33·4	21·7	119	+65	700	} eliminated, see Section 4.							
GB 9	929176	5	3	2·94	36·3	20·8	112	+72	1000								
(ii) Insch Mass																	
GN 1	728267	5	5	4·74	15·6	20·0	174	+24	300	6	5·96	111·7	6·4	159	+14	20	−23
GN 2	727267	5	6	5·34	7·6	25·9	159	+11	200	7	6·84	38·1	9·9	147	+7	34	−24
GN 3	734256	5	5	4·60	10·0	25·4	147	+37	900	4	3·63	8·2	34·2	146	+31	30	−7
GN 4	758267	5	7	6·93	80·8	6·8	138	+35	700	8	7·79	33·5	9·7	137	+31	39	−8
GN 5	801258	5	5	3·90	3·6	46·8	190	+32	700	5	4·80	20·0	17·6	167	+4	12	−30
GN 6	797258	5	6	5·64	13·9	18·6	161	+31	200	6	5·90	49·8	9·6	167	+15	12	−24
GN 7	737292	5	3	2·98	120·6	11·3	149	+33	300	3	2·99	291·1	7·2	146	+25	32	−14
GN 8	738289	5	5	4·83	23·5	16·1	113	+46	300	4	3·98	147·6	7·6	137	+26	40	−11
GN 9	748314	5	5	4·69	12·9	22·1	55	+68	600	5	4·34	6·1	33·8	121	+63	39	+23
GN 10	579247	7	7	4·30	2·2	33·6	107	+42	500	4	3·60	7·4	36·1	167	−9	13	−36
GN 11	625272	3	2	1·82	5·6	148·2	162	+36									
GN 12	652298	4	3	2·68	6·2	54·6	37	+81	300	4	3·92	39·2	14·9	128	+66	33	+24
GN 13	668327	3	2	1·86	7·4	111·9	164	+66	300	3	2·86	14·6	33·5	134	+34	41	−5

(iii) Boganclogh																
GC 1	425261	5	5	4·91	43·2	11·8	202	+53	300	5	4·95	74·8	8·9	198	+52	342	+1
GC 2	461278	5	6	5·25	6·7	28·0	210	+66	300	4	3·97	87·1	9·9	202	+38	336	−9
GC 3	474284	5	4	3·99	466·5	4·3	206	+52	200	4	3·99	258·9	5·7	202	+48	338	−1
(iv) Maud																	
GM 1	896510	6	6	5·96	123·0	6·1	164	+37	300	6	5·94	90·2	7·1	162	+35	15	−12
GM 2	919480	6	6	5·95	102·5	6·7	174	+46	300	5	4·97	156·8	6·1	173	+42	5	−8
(v) Arnage																	
GA 1	930375	5	5	4·95	80·4	8·6	175	+44	300	3	2·99	327·2	6·8	176	+36	1	−12
GA 2	933357	5	5	4·97	131·8	6·7	193	+40	300	5	4·96	89·9	8·1	193	+40	346	−9
GA 3	939328	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
(vi) Haddo House																	
GD 1	889360	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
(vii) Huntly																	
GH 1	540395	5	7	6·38	9·8	20·4	183	+63	300	6	5·41	8·5	24·3	206	+78	345	+36
GH 2	555425	6	8	7·35	10·8	17·6	182	+61	300	8	7·98	426·9	2·7	179	+48	358	−4
GH 3	456446	5	4						600	5	4·66	11·7	23·3	191	+35	347	−13
GH 4	492407	5	4	3·97	9·0	9·7	340	+66	700	4	3·70	10·1	30·4	139	+60	29	+14
GH 5	517408	6	8	4·60	2·1	52·8	5	+67	—	—	—	—	—	—	—	—	—
GH 6	501473	5	5	4·73	14·8	20·5	343	+83	300	2	1·90	9·5	93·0	142	+33	34	−8
GH 7	549478	5	2	1·95	18·9	61·0	206	+51	300	4	3·97	114·9	8·6	217	+43	324	−2
GH 8	593663	6	11	9·62	7·2	18·2	190	+51	—	—	—	—	—	—	—	—	—
GH 9	555528	6	6	5·32	7·4	26·4	58	+72	—	—	—	—	—	—	—	—	—
GH 10	554527	5	5	4·21	5·1	37·7	182	+61	—	—	—	—	—	—	—	—	—

* Each site mean direction includes one stepwise thermally demagnetized sample; the directions chosen were obtained after cleaning in various temperatures between 300 °C and 550 °C.

3.2 *Total NRM and susceptibility*

The cores have been treated in the laboratory as 2·5 cm long and 2·5 cm diameter cylinders. Magnetic moments have been measured using a PAR-SM2D spinner magnetometer and susceptibilities have been determined with a bridge of the kind described by Collinson & Molyneux (1967). Intensity (M_0) varies from 9×10^{-3} Gauss to 4×10^{-5} Gauss with a logarithmic mean 9×10^{-5} Gauss. Susceptibility (K) ranges from 8×10^{-6} to 5×10^{-3} Gauss/Oersted with logarithmic mean 4×10^{-5} Gauss/Oersted. The bulk of the Q_n ratios ($= M_0/K$) lie in the range 0·5–1·5 which is typical of palaeomagnetic material with stable NRM (Irving 1964, page 92). Of the five sites with Q_n less than 0·3 (GA3, GN11, GN12, GN13, GH6) only the first one has non-significant NRM directions before cleaning and the first two become non-significant after cleaning; the others give well-grouped directions close to the remainder of the collection and it is inferred that they were either magnetized in a weak field or have experienced viscous decay of their moments since formation. As a rule the grouping of NRM directions shows moderate precision within each site and only four are not significant at the 95 per cent confidence level (Irving 1964, page 62).

3.3 *AF demagnetization results*

One specimen from each site has been progressively demagnetized by AF treatment, and the majority show an exponential fall-off in their moments with very little change in the direction of the NRM (e.g. GN2b in Fig. 2 (a)). Such specimens with a broad coercivity spectrum probably possess a wide range of domain sizes (Stacey 1963). Small secondary components of IRM were present in some of the pilot samples (GB3, GB6, GB7, GN4, GN6, GN13) and in most cases are eliminated after treatment in fields less than 500 Oe. These soft components are identified by a sharp fall in the magnetic moment (Fig. 2(c)) coupled with a large change in the NRM direction (Fig. 2(c)) coupled with a large change in the NRM direction (Fig. 2(a)). Once the soft components have been removed the magnetic moments decrease gradually upon further demagnetization as does the bulk of the collection, and little or no change in direction is observed. A few specimens (GB4, GH5, GA3, GH8, GH10) behave in an unstable fashion. Two specimens (GN13, GH2) show very little change in their magnetic moments upon AF demagnetization (Fig. 2(b) and (d)): the first shows a decrease of 12 per cent after demagnetizing in fields of 1200 Oe, and the second a decrease of 23 per cent after demagnetizing at 1000 Oe (Fig. 2(d)). Such high coercivities are usually attributed to haematite but Evans, McElhinny & Gifford (1968) who have observed high coercivities in similar rocks, concluded that they may originate in exsolved magnetite crystals within ilmenite or pyroxenes rather than in the primary magnetite.

In polished section fine elongate crystals of titanomagnetite in parallel groups have been observed, exsolved along the prismatic cleavages of pyroxenes, and it is suggested that these may carry the component of high coercivity. Elongate magnetite can have coercivity of several thousand Oersteds if the axial ratios exceed 1 : 10 when they will act as single domain grains (Morrish & Yu 1955; Strangway, Larson & Goldstein, 1968).

The optimum field for AF cleaning of the remainder of the collection has been determined for each site from the behaviour of the pilot using Briden's (1972) stability index. The fields vary between 100 and 1000 Oe but are mostly below 700 Oe. The within-site grouping is improved for 22 of the original 41 sites after AF cleaning (Table 1). Five sites (GN11, GH8, GH10, GH5, GH9) are no longer statistically significant at the 95 per cent confidence level after cleaning. The first three

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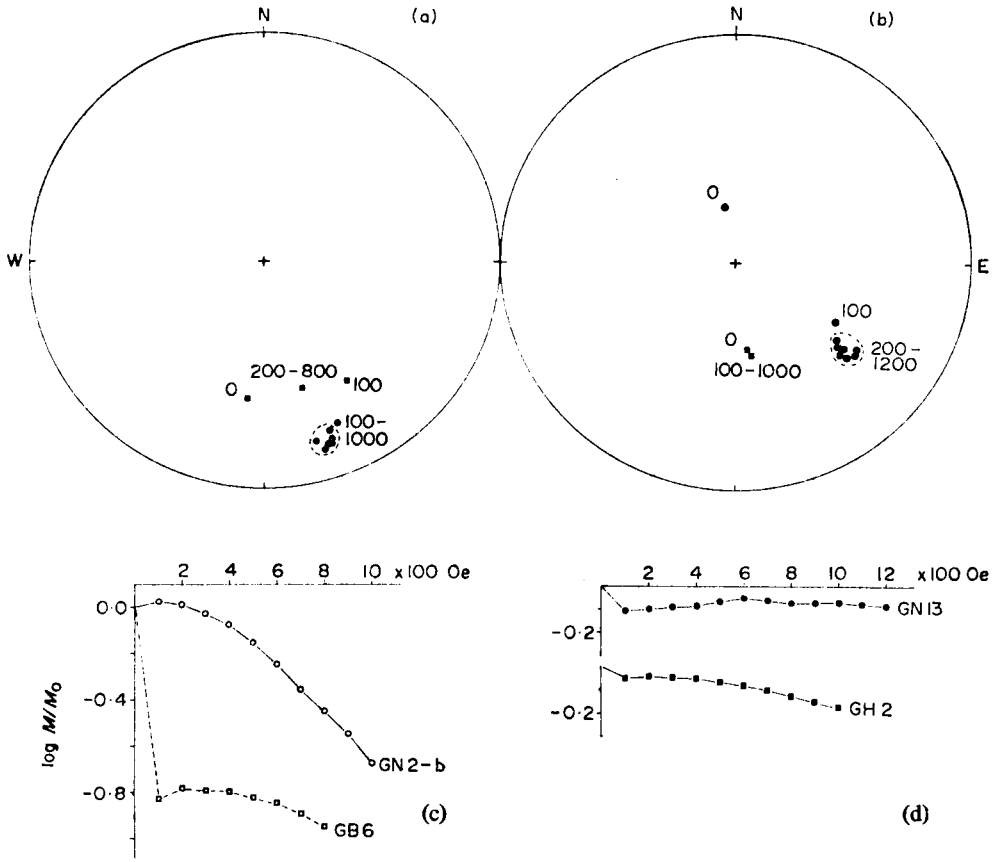


FIG. 2. Change of remanence directions on AF demagnetization of (*a*) a gabbro sample (GN 2—b) showing approximately an exponential fall in moment and a sample (GB 6) showing an IRM component, and (*b*) two samples (GH 2, GN 13) showing only slight response to AF treatment, one of which shows a small IRM component. (*c*) and (*d*) show the change in moment on demagnetization of the samples illustrated in (*a*) and (*b*). In the stereograms of Figs 2, 3 and 4 open symbols denote upwards, closed symbols downwards.

of these had total NRMs close to the stable *in situ* direction typical of the gabbro bodies, and have been excluded from the analysis only because the stability index is regarded here as the main criterion for stability.

3.4 *Thermal demagnetization*

One or two specimens from each site have been demagnetized in several steps up to 685 °C. Most show little or no change in NRM direction in the course of treatment, and those which do tend to approach the directions obtained by AF demagnetization (Fig. 3(a)). A sharp fall in all demagnetization curves is observed in the range 500–600 °C (Fig. 3(b)). In 29 of the 35 specimens, less than 10 per cent of the original moment is detectable after treating in temperatures of 600 °C and this remanence is attributed to minerals with high blocking temperatures and possibly oxidation of the magnetic minerals during heating. Coverage of the interval 530–680 °C in steps of about 50 °C showed random changes in direction.

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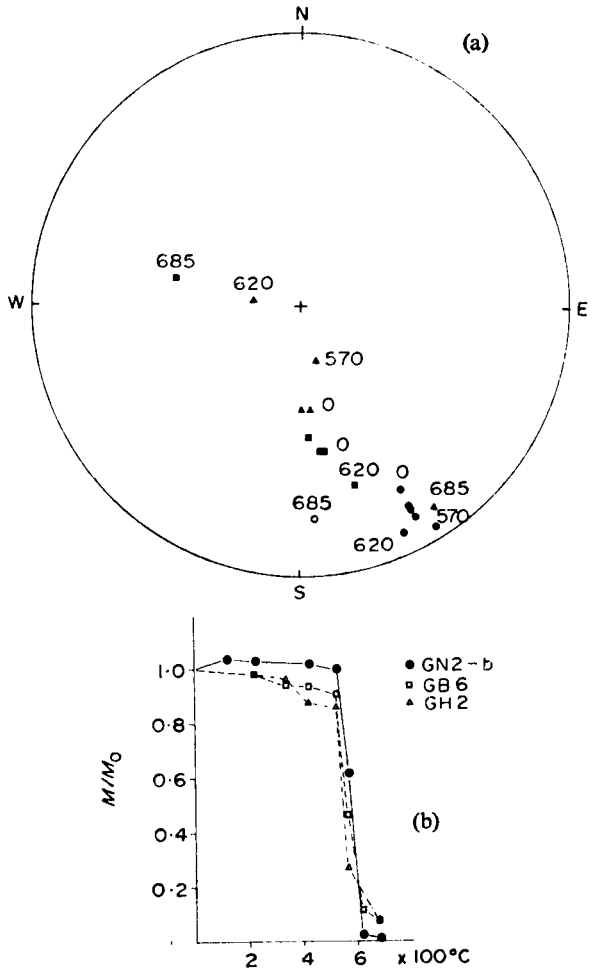


FIG. 3. Thermal demagnetization of three pilot specimens showing (*a*) change in remanence directions, and (*b*) change in moment.

4. Analysis and interpretation

Since there is no appreciable difference between directions after AF and thermal cleaning, the two are combined in Table 1 for the computation of individual site means. Thirty four statistically significant site mean directions are well grouped, five sites having failed to respond to cleaning and two being random before and throughout treatment. Another two sites (B8, B9) are eliminated from further discussion because anomalous remanence directions reinforce earlier doubts whether their outcrops were *in situ*.

All the rocks have the same polarity, with southerly declinations and positive inclinations, as was found by Blundell & Read (1958); statistics are summarized in Tables 2 and 3.

A fold test is applied to the Huntly, Insch and Belhelvie masses to determine the age of the magnetization relative to the primary igneous structures as observed by Munro (1970) for the Huntly mass, Clarke & Wadsworth (1970) for the Insch mass and Stewart (1946) for the Belhelvie mass. The dips in the Huntly and Belhelvie

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masses are near vertical, and shallow in the Insch mass. For both the Insch and Huntly masses (Table 3) the precision parameter k is significantly higher *before* allowing for the dip, using the test of McElhinny (1964) and Cox (1969), and implies that the magnetization was acquired with the lamination in its present attitude. The tilting of the igneous lamination is associated with the formation of the major open folds in the area and is referred to phase F_3. It follows that the remanence post-dates F_3. This observation is supported by the fact that the mean directions in the three masses are more different after allowing for the dip (Fig. 4); those of the

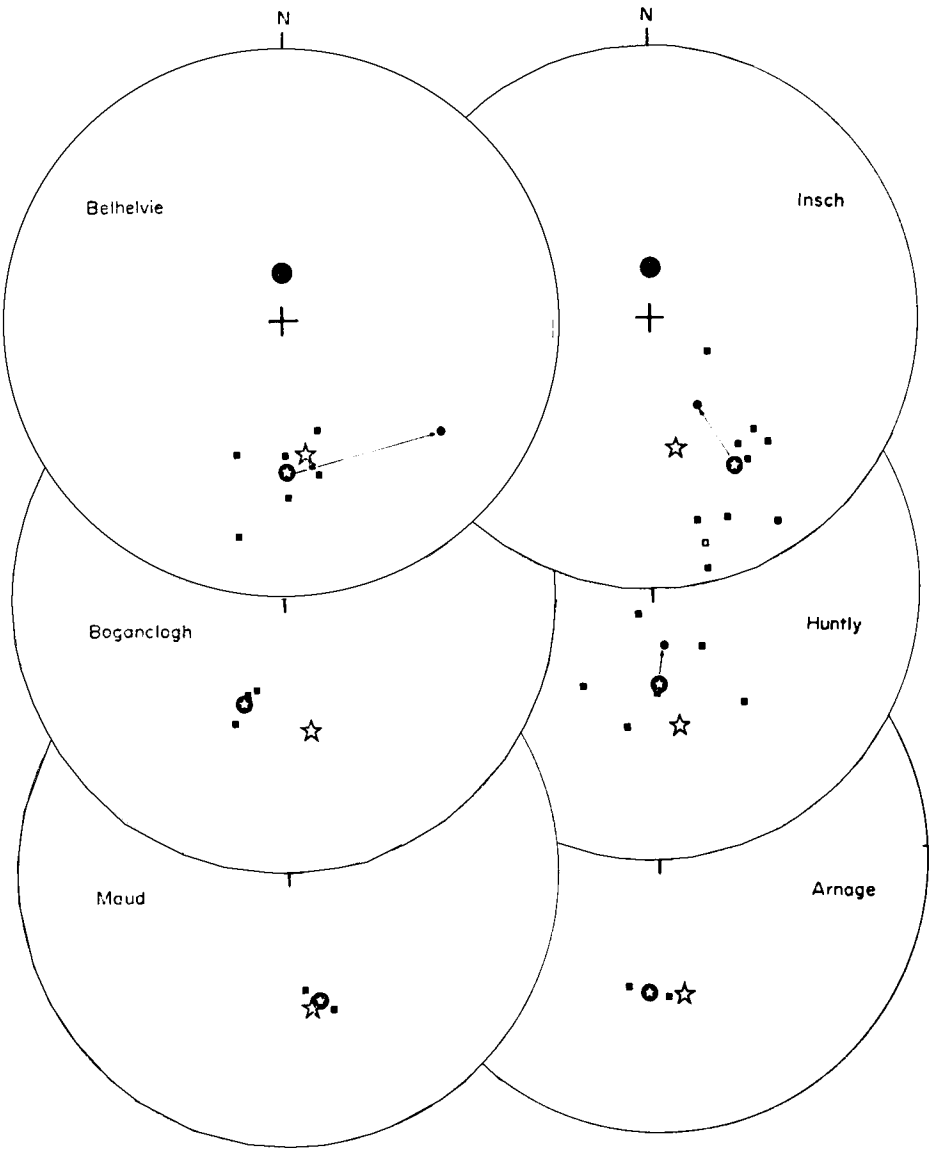


FIG. 4. *In situ* directions of stable magnetization in each mass of the Younger Gabbros. The large star is the overall mean direction of magnetization of all the 32 sites and the solid circle is the direction of the present geocentric axial dipole field. The encircled star is the *in situ* mean for each mass and for Belhelvie, Insch and Huntly the mean after rotating igneous lamination to the horizontal is denoted by a smaller solid circle.

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Table 2

Mean in situ *palaeomagnetic results for each gabbro mass before and after AF and thermal cleaning*

	Total NRM								After cleaning								
							Palaeomagnetic North Pole								Palaeomagnetic North Pole		
	N (sites)	R	k	α_{95}	D	I	°E	°N	N (sites)	R	k	α_{95}	D	I	°E	°N	(dψ, dχ)
Belhelvie	8	6·47	4·6	29·1	150	+76	13	+32	7	6·78	27·3	11·7	180	+32	358	−15	
Insch	13	11·40	7·4	16·3	149	+46	25	−1	12	10·90	10·0	14·4	149	+26	28	−14	
Bogancloch	3	2·98	103·8	12·2	206	+57	337	+7	3	2·98	112·2	11·7	201	+46	339	−4	
Huntly	9	8·13	9·5	17·9	190	+79	353	+36	6	5·48	9·6	22·8	177	+53	0	+1	
Maud	2	1·99	92·3	26·3	169	+41	8	−8	2	1·99	111·0	23·9	167	+39	10	−10	
Arnage	2	1·99	67·9	30·8	184	+42	354	−8	2	1·99	72·2	29·8	184	+38	354	−11	
Combined	37	31·2	6·3	10·3	164	+62	10	+11	32	28·8	9·7	8·6	168	+34	9	−11	(6, 10)
Blundell & Read (1958)	21 (samples)	—	13	9	182	+51	356	−1									(8, 12)

Table 3

'Fold test' applied to the igneous lamination in the Belhelvie, Insch and Huntly gabbro masses

	N	R	k	α_{95}	D	I
(i) *In situ*						
Belhelvie	7 sites	6·78	27·4	11·8	180	+32
Insch	12 sites	10·90	10·0	14·4	149	+26
Huntly	6 sites	5·48	9·6	22·8	177	+53
Combined	25 sites	22·26	8·8	10·4	163	+35
	3 masses	2·88	17·3	30·6	167	+38
(ii) Corrected for dip of igneous lamination						
Belhelvie	7	6·80	30·3	11·1	123	+20
Insch	12	7·90	2·7	33·1	149	+48
Huntly	6	4·02	2·5	53·4	172	+65
Combined	25 sites	17·45	3·2	19·5	139	+43
	3 masses	2·77	8·6	44·8	142	+46

Huntly and Belhelvie masses are much further from the overall mean than they are *in situ*. The *in situ* mean for the Insch mass is slightly anomalous compared with the other bodies, and when the igneous lamination is restored to horizontal, the direction for this mass is brought closer to the direction of the other masses (Fig. 4). This might be taken as evidence in support of the view of Fettes (1970) that the Insch mass was tilted by the forceful upward intrusion of the Bennachie granite (one of the Younger Granite suite, about 400 My in age). However, the Boganclogh mass is a westward extension of the Insch mass, but its direction of magnetization deviates from the mean in the opposite sense to the anomaly of the Insch mass whether or not a 'dip correction' is applied to the Boganclogh mass. It may be, then, that neither of these palaeomagnetic discrepancies is ascribable to tectonic tilting. Thus, with the possible exception of the Insch mass, the stable remanence throughout the gabbros is most consistent when viewed *in situ*; this implies that the remanence was acquired when the rocks were in their present position. Hence if the primary igneous structures define the horizontal at the time of crystallization, then their folding took place before they were magnetized. Since the AF and thermal demagnetization results, and Q_n ratios strongly suggest that the magnetization is primary, the deformation of the igneous lamination must have occurred above the Curie temperatures in these rocks, which approach 600 °C. This substantiates earlier arguments of Stewart & Johnson (1960) and Wadsworth, Stewart & Rothstein (1966).

There is chemical evidence for high pressures during the crystallization of the gabbros (Gribble 1967) and the partial fusion associated with them implies temperatures of 600 °C or possibly more at the time of their intrusion together with pressures of a few kilobars (Pankhurst 1970). It is difficult to estimate the precise age of magnetization. It must be intermediate between the Rb-Sr age 486 ± 17 My identified as the time of crystallization by Pankhurst (1970) and the K-Ar age of 461 ± 5 My identified by him with the time of cooling through about 200 °C. The gabbros are believed to have been intruded near the climax of the main (M_3) metamorphism in the region and it is unlikely that any subsequent discrete event reached 600 °C. Hence a long slow cooling history is envisaged. In the absence of direct information we can do no better than assume uniform cooling from (say) 1200 °C at 486 My to 200 °C at 461 My. This yields an estimate of 470 My for the time of cooling through the Curie temperature, which receives some support from an inference of Pankhurst (1970) that hornblendes and pyroxenes in some of these

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rocks began closing to argon at about that time. The best available estimate of the palaeomagnetic field direction at that time is regarded as the *in situ* mean computed from the 32 grouped site mean directions after AF and thermal cleaning, i.e. $D = 168°$, $I = +34°$ which corresponds to a palaeomagnetic pole at 9° E and 11° S. (If the possible post-magnetization tilting of the Insch mass were taken into account, the mean direction and mean pole would only be altered by about five degrees.) This computation is summarized in Table 2 together with the result of Blundell & Read (1958). The present result is preferred because of the more extensive sampling, which enables unit weight to be given to sites instead of samples in the computation, and the detailed stability tests. By comparison with the earlier study, the precision and cone of 95 per cent confidence are only slightly changed but the mean direction is altered by 20° and the pole by 16°.

In conclusion we would modify the contention of Blundell & Read (1958) that the gabbros have not been tectonically disturbed since their *formation*; we merely argue that (with the possible exception of the Insch mass) they have not been disturbed since they were *magnetized* and infer that the F_3 deformation occurred during their long post-emplacement cooling interval ($\sim 486 \pm 17$ to 470 My, say).

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